

8/7/20 H-W

1. Define a structure in C and Cpp 2.

The structure should store the variables for student type. Compute the size of the structure manually.

2. Use size of function and display the size. Cross check if both are same or not.

eg. Declare self ref structures

i) emp

```
struct emp
```

```
{
```

```
int eid;
```

```
char ename[50];
```

```
struct emp *next;
```

```
}
```

ii) student

```
struct student
```

```
{
```

```
int rno;
```

```
char name[45];
```

```
struct student *next;
```

```
}
```

iii) node.

```
struct node
```

```
{
```

```
int data;
```

```
float data2;
```

```
struct node *ptr;
```

```
}
```

HW

i) Using C

```
#include <stdio.h>
```

```
struct student {
```

```
int rollno; // 4 bytes
```

```
char grade; // 1 byte
```

```
float marks; // 4 bytes
```

```
};
```

```
int main() {
```

```
struct student s;
```

```
printf("Size of structure using sizeof(): %lu  
bytes\n", sizeof(s));
```

```
return 0;
```

```
}
```

output: Size of structure using sizeof(): 12 bytes

using C++:

```
#include <iostream>
```

```
using namespace std;
```

```
struct student {
```

```
int rollno; // 4 bytes
```

```
char grade; // 1 byte
```

```
float marks; // 4 bytes
```

```
};
```



```

int main() {
    student s;
    cout << "size of structure using sizeof() is " << "12 bytes"
    << endl;
    return 0;
}

```

Output: Size of structure using sizeof() : 12 bytes

Manual size calculation : 9 bytes } They are not same.
 using sizeof() : 12 bytes

Justification:

The manual size and the size returned by sizeof() is not same manually because of "Structure padding" which is known as "memory Alignment".

generally, int = 4 bytes

char = 1 byte

float = 4 bytes

however the compiler stores like int = 4 bytes, char = 1 byte, padding = 3 bytes, float = 4 bytes, so they are not same because compiler adds padding bytes to align the data members in memory. It improves the performance of memory access.

Recursion:

Sum of n numbers:

```

int sum(int n)
{
    if (n == 1)
        return 0;
    else
        return n + sum(n-1);
}

```